

Tidal disruption of the LMS-1/Wukong progenitor with two embedded globular clusters and dynamical friction: methods and orbit–anchor diagnostics

1 Aim

Yuan et al. (2020) and Malhan et al. (2021) associate the globular clusters NGC 5024 (M53) and NGC 5053 with the LMS-1/Wukong accretion event, but their orbital models contain *no dynamical friction*. We rewind a progenitor dwarf in the Milky Way (MW) potential, plant the two clusters inside it, and integrate forward to the present with and without Chandrasekhar friction. We then confront the resulting stellar debris with the observed LMS-1 stream from the STREAMFINDER catalogue (Ibata et al. 2024), and we test how the result depends on the progenitor halo mass, the disruption time, and crucially the choice of orbit on which the debris is launched.

2 Galactic frame and host potential

All dynamics are integrated in the Milky Way potential of Bovy (2015, `MWPotential2014`), $\Phi_{\text{MW}}(\mathbf{x})$. Observed equatorial phase space is mapped to the Galactocentric frame with $R_0 = 8 \text{ kpc}$, $z_\odot = 25 \text{ pc}$, and solar velocity $\mathbf{v}_\odot = (11.1, 232.24, 7.25) \text{ km s}^{-1}$, and back, using the standard ICRS $\leftrightarrow$ Galactocentric transformation. We denote a 6D state $\mathbf{w} = (\mathbf{x}, \mathbf{v})$.

3 LMS-1 progenitor dark halo

The progenitor is an NFW halo whose structural parameters follow the Dutton & Macciò (2014) concentration–mass relation at formation redshift $z_h = 1$. For a virial mass M_{200} ,

$$\log_{10} c_{200} = a + b \log_{10} \left(\frac{M_{200} h}{10^{12} M_\odot} \right), \quad (1)$$

$$a = 0.520 + (0.905 - 0.520) e^{-0.617 z^{1.21}}, \quad b = -0.101 + 0.026 z, \quad (2)$$

$$\rho_{\text{crit}}(z) = \rho_{\text{crit},0} E^2(z), \quad E^2(z) = \Omega_m (1+z)^3 + \Omega_\Lambda, \quad (3)$$

$$r_{200} = \left(\frac{3M_{200}}{4\pi 200 \rho_{\text{crit}}(z)} \right)^{1/3}, \quad r_s = \frac{r_{200}}{c_{200}}, \quad (4)$$

$$M_c = \frac{M_{200}}{\ln(1 + c_{200}) - c_{200}/(1 + c_{200})}, \quad (5)$$

with $h = 0.70$, $\Omega_m = 0.30$, $\Omega_\Lambda = 0.70$. The NFW density and potential are

$$\rho_{\text{NFW}}(r) = \frac{M_c}{4\pi r_s^3} \frac{1}{(r/r_s)(1 + r/r_s)^2}, \quad \Phi_{\text{NFW}}(r) = -\frac{GM_c}{r} \ln \left(1 + \frac{r}{r_s} \right). \quad (6)$$

The stellar body and each cluster are softened Plummer spheres,

$$\Phi_{\text{P}}(r) = -\frac{GM}{\sqrt{r^2 + b^2}}, \quad (7)$$

with scale $b_{\text{LMS1}} = 2 \text{ kpc}$ for the stars and $b_{\text{GC}} = 10 \text{ pc}$ for the clusters. The internal circular speed and 1D velocity dispersion of the dwarf are

$$v_{\text{circ}}(r) = \sqrt{r |\partial_r \Phi_{\text{LMS1}}|}, \quad \sigma(r) = \max\left(\frac{v_{\text{circ}}(r)}{\sqrt{2}}, \sigma_{\text{floor}}\right), \quad (8)$$

where $\Phi_{\text{LMS1}} = \Phi_{\text{NFW}} + \Phi_{\text{P, stars}}$.

4 Chandrasekhar dynamical friction

A body of mass M_{sat} moving with velocity $\mathbf{v}_{\text{rel}}$ relative to a background of density ρ and dispersion σ feels the drag

$$\mathbf{a}_{\text{DF}} = -\frac{4\pi G^2 M_{\text{sat}} \ln \Lambda \rho}{v_{\text{rel}}^3} \left[\text{erf}(X) - \frac{2X}{\sqrt{\pi}} e^{-X^2} \right] \mathbf{v}_{\text{rel}}, \quad X = \frac{v_{\text{rel}}}{\sqrt{2} \sigma}. \quad (9)$$

Equation (??) is applied in two places:

- *halo–MW friction* on the dwarf’s orbit, with $\rho = \rho_{\text{MW}}$, $\sigma = \sqrt{\max(-\Phi_{\text{MW}}/2, \sigma_{\text{min}}^2)}$, $M_{\text{sat}} = M_{200}$, $\ln \Lambda = 4.6$;
- *cluster–halo friction*, with $\rho = \rho_{\text{NFW}}$, $\sigma = \sigma(r)$, $M_{\text{sat}} = M_{\text{GC}}$, $\ln \Lambda = \max[2, \ln(M_{200}/M_{\text{GC}})]$.

The “no-DF” runs set $\mathbf{a}_{\text{DF}} \equiv 0$ everywhere, reproducing the Yuan/Malhan assumption.

5 Time integration

All trajectories use a kick–drift–kick (KDK) leapfrog with fixed step $\tau = 2^{-11} \text{ Gyr} \approx 0.49 \text{ Myr}$:

$$\begin{aligned} \mathbf{v}_{n+1/2} &= \mathbf{v}_n + \frac{\tau}{2} \mathbf{a}(\mathbf{x}_n, \mathbf{v}_n), \\ \mathbf{x}_{n+1} &= \mathbf{x}_n + \tau \mathbf{v}_{n+1/2}, \\ \mathbf{v}_{n+1} &= \mathbf{v}_{n+1/2} + \frac{\tau}{2} \mathbf{a}(\mathbf{x}_{n+1}, \mathbf{v}_{n+1/2}). \end{aligned} \quad (10)$$

Backward integration (“rewind”) uses $\tau \rightarrow -\tau$.

5.1 Equations of motion

The moving halo centre, the (massless) stream tracers, and the live clusters obey

$$\mathbf{a}_{\text{centre}} = -\nabla \Phi_{\text{MW}}(\mathbf{x}_c) + \mathbf{a}_{\text{DF}}^{\text{halo}}, \quad (11)$$

$$\mathbf{a}_{\text{star}} = -\nabla \Phi_{\text{MW}}(\mathbf{x}) - \nabla \Phi_{\text{NFW}}(\mathbf{x} - \mathbf{x}_c) - \sum_i \nabla \Phi_{\text{GC},i}(\mathbf{x} - \mathbf{x}_{\text{GC},i}), \quad (12)$$

$$\mathbf{a}_{\text{GC},i} = -\nabla \Phi_{\text{MW}}(\mathbf{x}_i) - \nabla \Phi_{\text{NFW}}(\mathbf{x}_i - \mathbf{x}_c) - \sum_{j \neq i} \nabla \Phi_{\text{GC},j}(\mathbf{x}_i - \mathbf{x}_j) + \mathbf{a}_{\text{DF},i}^{\text{halo}}, \quad (13)$$

where $\mathbf{x}_c(t)$ is the rewind halo trajectory (below) and the stream tracers are massless.

6 Progenitor orbit by rewinding the clusters

The present-day LMS-1 centroid is the *mass-weighted* mean of the two clusters' measured phase space,

$$\mathbf{w}_{\text{LMS1}}(0) = \frac{M_{5024} \mathbf{w}_{5024} + M_{5053} \mathbf{w}_{5053}}{M_{5024} + M_{5053}}. \quad (14)$$

Mass-weighting is required: a plain average cancels the clusters' opposite v_{los}/v_z and yields an unphysically circular orbit. This state is rewound for T in Φ_{MW} (+ $\mathbf{a}_{\text{DF}}^{\text{halo}}$ for the DF run) to give $\mathbf{x}_c(t)$, $t \in [-T, 0]$, the moving NFW centre.

7 Cluster initial conditions (corrected)

Rather than planting the clusters at arbitrary internal offsets, we set their $t = -T$ states by rewinding their *observed* present-day phase space $\mathbf{w}_{\text{GC},i}(0)$ through the full force field (MW + moving NFW + mutual gravity [+DF]). Forward integration then returns each cluster to its observed position: exactly for the reversible no-DF case, and to $\lesssim 0.1^\circ$ for the (velocity-dependent, weakly irreversible) DF case. A by-product is that, on its measured orbit, NGC 5053 lies ~ 89 kpc from the LMS-1 centre at $t = -5$ Gyr (versus ~ 3 kpc for NGC 5024), i.e. it is not tightly co-orbiting the core.

8 Stellar stream sampling

$N = 10^4$ massless tracers are drawn from a quasi-spherical distribution function $f(\mathbf{x}, \mathbf{v})$ self-consistent with the Plummer stellar density inside Φ_{LMS1} , then rigidly translated to the rewound centre $\mathbf{x}_c(-T)$ and integrated forward, feeling MW + moving NFW + both clusters.

9 Tidal (Jacobi) radius

The instantaneous tidal radius of the dwarf at galactocentric distance R is solved self-consistently from the enclosed masses,

$$r_t = R \left[\frac{M_{\text{NFW}}(< r_t)}{3 M_{\text{MW}}(< R)} \right]^{1/3}, \quad M_{\text{NFW}}(< r) = M_c \left[\ln \left(1 + \frac{r}{r_s} \right) - \frac{r/r_s}{1 + r/r_s} \right], \quad (15)$$

with $M_{\text{MW}}(< R) = v_c^2(R) R/G = |a_R| R^2/G$.

10 Observable comparison

Final 6D debris is transformed to $(\alpha, \delta, d, \mu_{\alpha*}, \mu_\delta, v_{\text{los}})$. We select catalogue LMS-1 members, build the convex hull of their (α, δ) (dilated 15%) as a footprint $\mathcal{F}$, and compare only model particles inside $\mathcal{F}$. Coverage is the fraction of observed stars with ≥ 5 model neighbours within 3° ; kinematic agreement uses the median offsets and the local RMS residuals in μ and v_{los} .

To test whether the debris is launched on the right orbit, we measure the great-circle distance of each observed star to the projected progenitor orbit $\{\alpha(t), \delta(t)\}$,

$$\theta_k = \min_t \arccos(\hat{\mathbf{n}}_k \cdot \hat{\mathbf{n}}_{\text{orb}}(t)), \quad (16)$$

together with the angle between the on-sky stream direction and the local orbit tangent.

11 Stream-fit (Malhan) orbit

For the literature comparison we also build an IC that matches the Malhan et al. (2021) orbit ($r_{\text{peri}} \approx 10$, $r_{\text{apo}} \approx 16$ kpc, $e \approx 0.2$, $L_z \approx -700$ kpc km s $^{-1}$, inclination $\sim 75^\circ$). Anchoring at apocenter along the observed stream centroid, we solve for the tangential speed and orientation that reproduce $(r_{\text{peri}}, r_{\text{apo}}, L_z)$, verifying by direct integration.

12 Experiments and key results

- **DF vs. no-DF.** With DF both clusters sink to and are retained in the core; the rewind centre orbits differ from the no-DF case by up to ~ 86 kpc at -5 Gyr (but negligibly over the last $\sim$ Gyr, since both share the identical present-day state).
- **Disruption time** $T = 5\text{--}9$ Gyr. The footprint kinematics are essentially unchanged and the bound fraction drops only mildly with T : time is not the controlling parameter.
- **Halo mass** $M_{200} = 10^8\text{--}10^{11} M_\odot$. This is the decisive lever. Light haloes ($\sim 10^8$) fully disrupt into a long, cold stream reaching the observed $\alpha \approx 270^\circ$ with the best proper-motion match but zero bound core; heavy haloes ($\gtrsim 10^{10}$) retain the cluster pair but give a compact, under-extended blob. No single mass both makes the stream and retains the clusters.
- **Orbit anchor.** The cluster-derived orbit lies a median 16° off the observed stream (on-sky directions differ by 53°); debris launched on it never develops the observed μ_δ gradient. The Malhan stream-fit orbit follows the data to a median 0.7° ; disrupting the same dwarf on it yields coverage 66%, $\alpha_{\text{max}} \approx 270^\circ$, and footprint $\mu_\delta = -2.37$ vs. observed -2.34 mas yr $^{-1}$. The mismatch was the launch orbit, not dynamical friction.